



DIPLOMA IN CYBERSECURITY & DIGITAL FORENSICS (LEVEL 2)

FORENSICS IN DIGITAL SECURITY (CCD2C04)

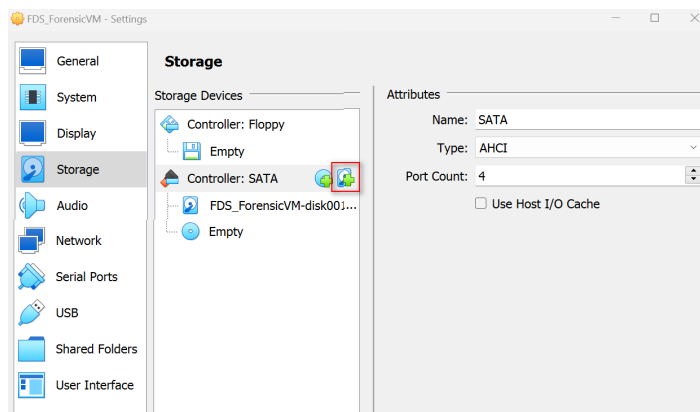
Practical 5 Acquisition

You will use Kali Live CD to perform several forensic tasks. The file name is **kali-linux-2025.1c-live-amd64**

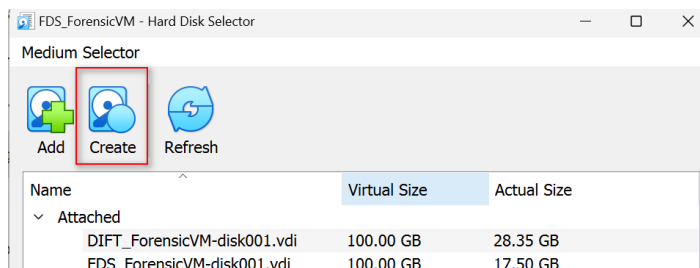
You would acquire a 200MB hard disk image and then restore it to a 300MB hard disk.

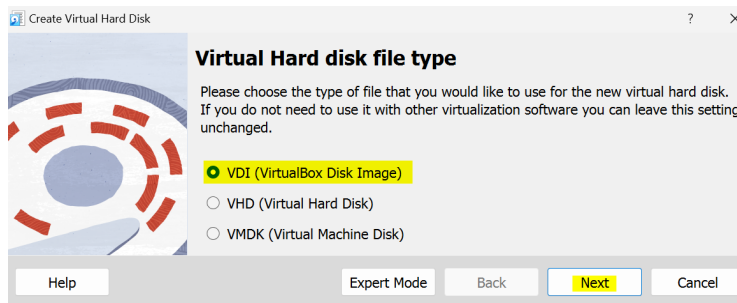
1. Part 1. Preparing the Hard disk

- a) In your **VirtualBox Windows VM**, create a 200MB Hard disk.
[Note: The VM must be powered off]

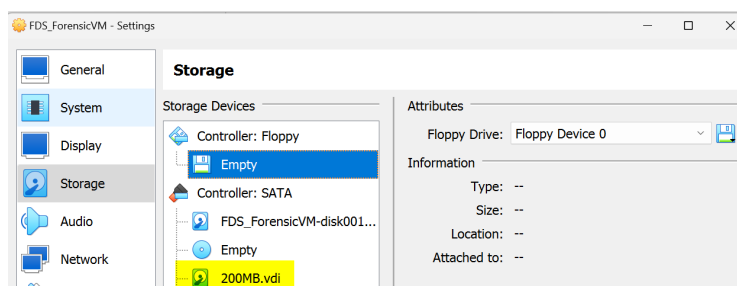
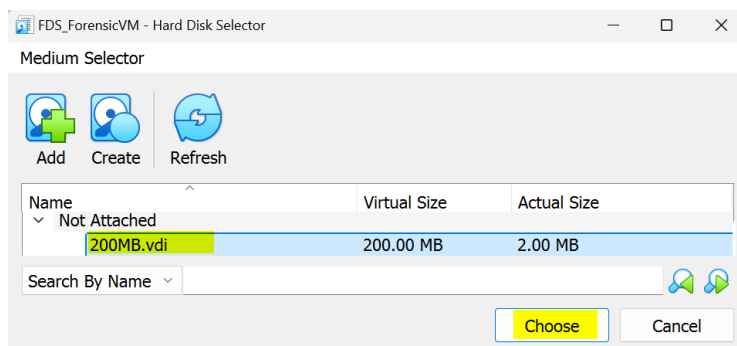
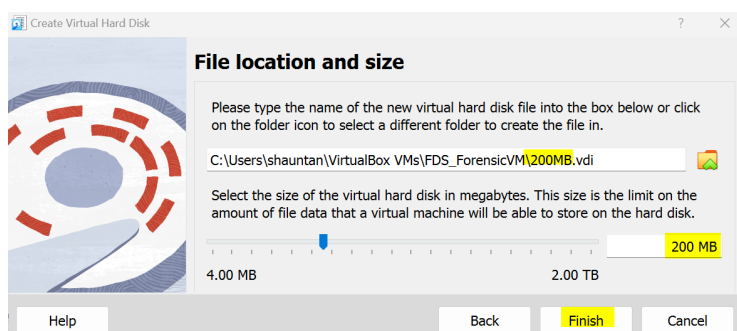


- b) In the **Create Virtual Hard Drive** dialog, select **VDI(VirtualBox Disk Image)** that is **Dynamically allocated** by default.

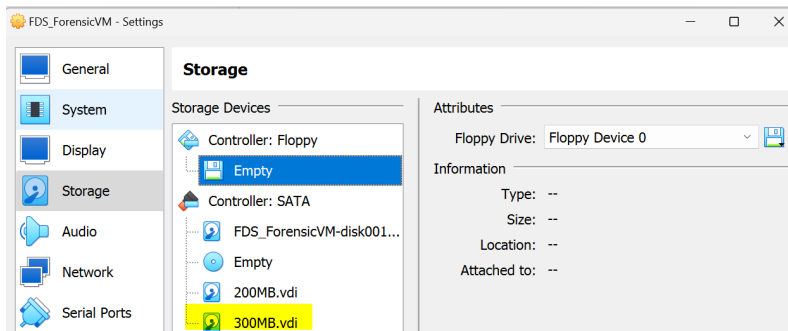




- c) Name the hard disk image as “**200MB.vdi**” and set it to 200MB. Finally, click the **Finish** button.

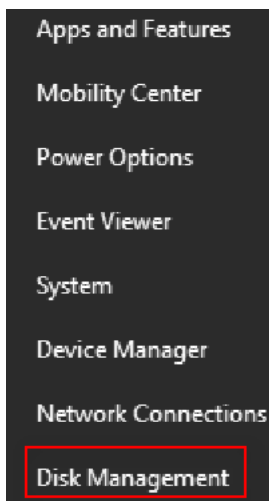


d) Repeat steps (a) to (c) to create a 300MB hard disk image.

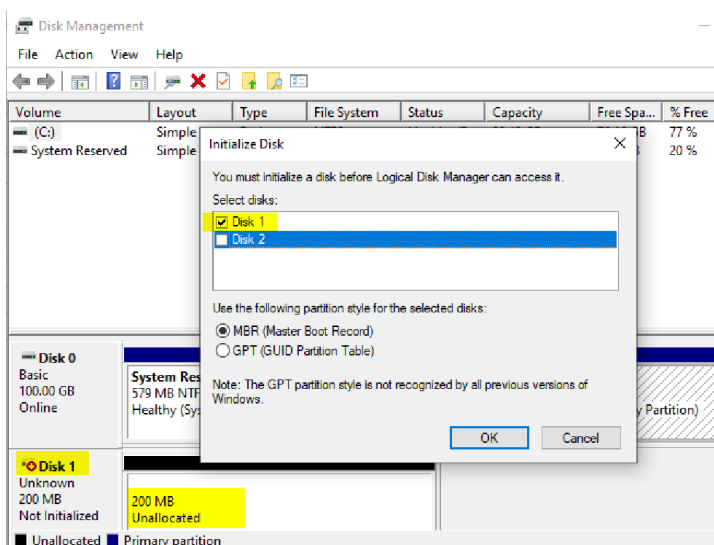


e) Start the **VirtualBox Windows VM**.

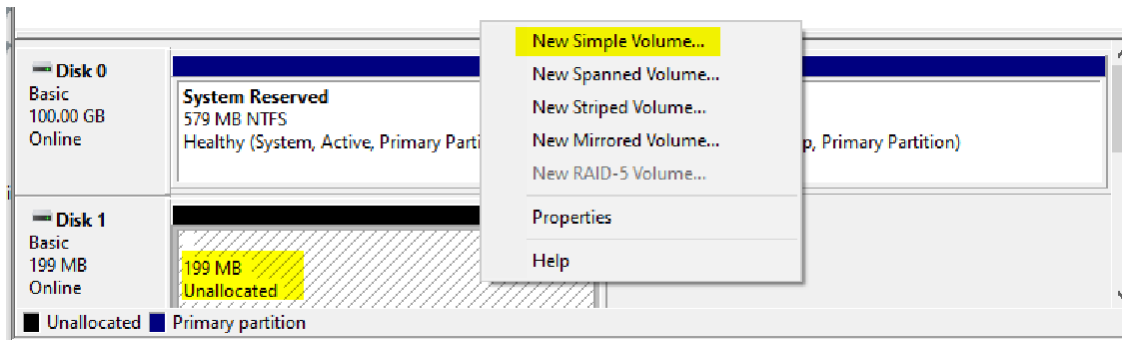
f) Initialize and format the new 200MB hard disk created using **Window Start > Disk Management**.



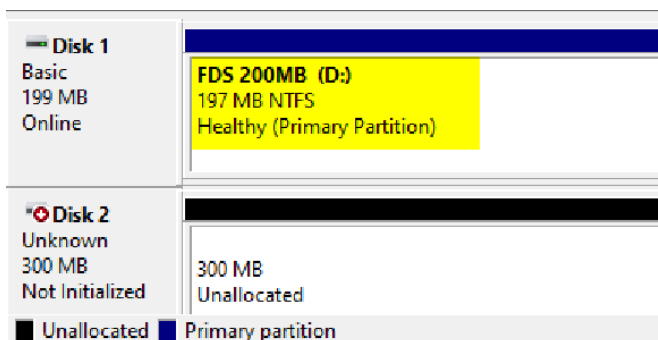
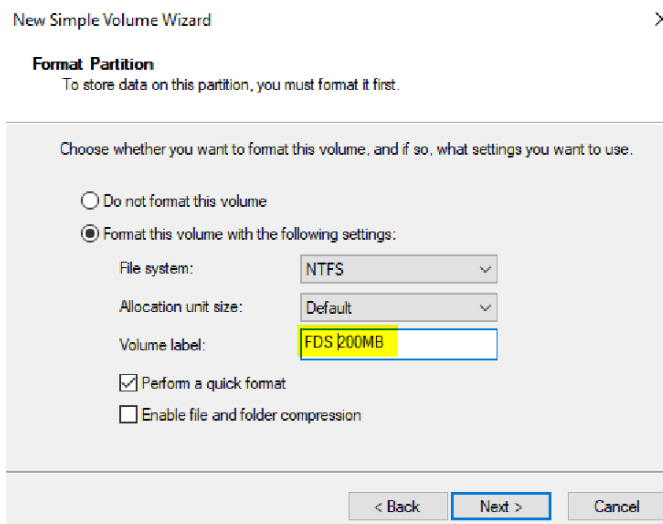
g) When prompted with **Initialize Disk** dialog, select only Disk 1 and accept all default configurations.



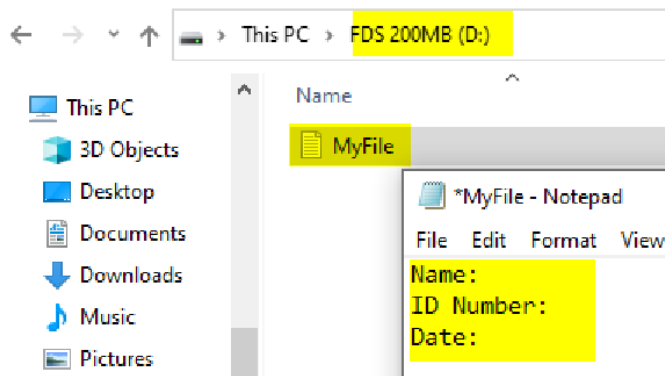
- h) Create a new volume by right-clicking on the unallocated 200MB disk shaded area > **New Simple Volume**. Accept all default configurations.
 [Note: DO NOT create a volume on the 300MB hard disk.]



- i) Format the new 200MB hard disk by right-clicking on the 200MB disk > **Format** with all default settings. Set the **Volume Label** to be "FDS 200MB" and accept all other default configurations. [Note: DO NOT format 300MB hard disk.]

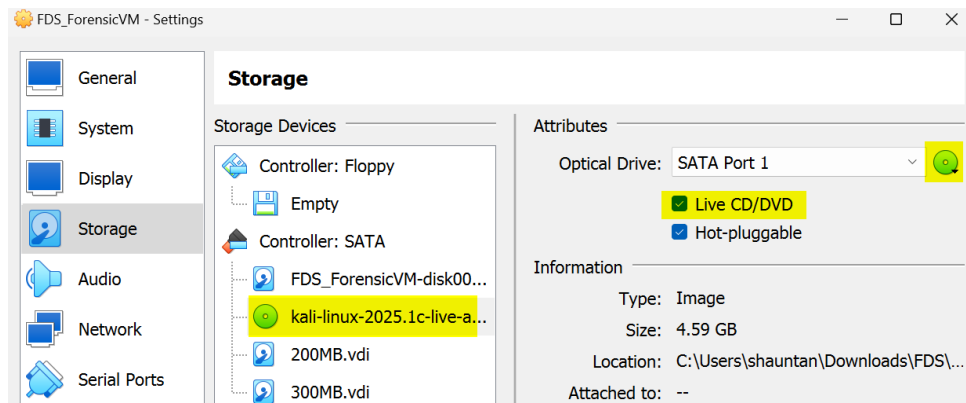


- j) Use **Notepad**, create a simple text file that contains your name, matric number, and place a message in it. Save this file on the new 200MB hard disk.



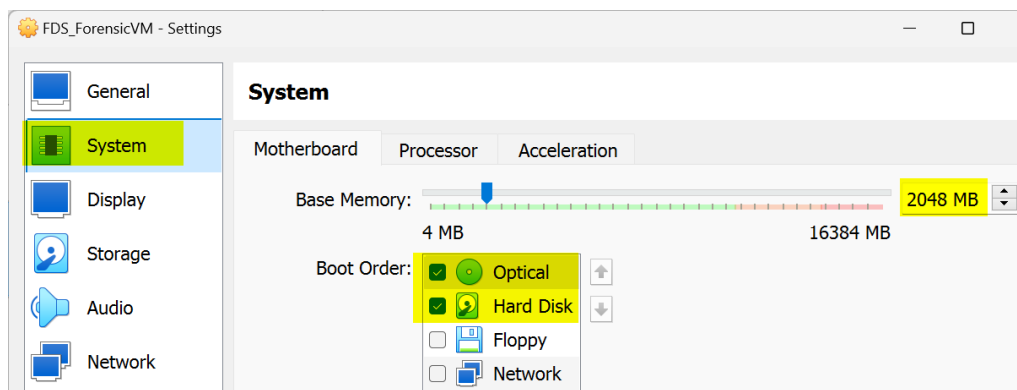
2. Part 2. Using Kali for Bit-Stream Extraction

- a) Shut down the VM and attach the **Kali ISO** to the **Windows VM**.

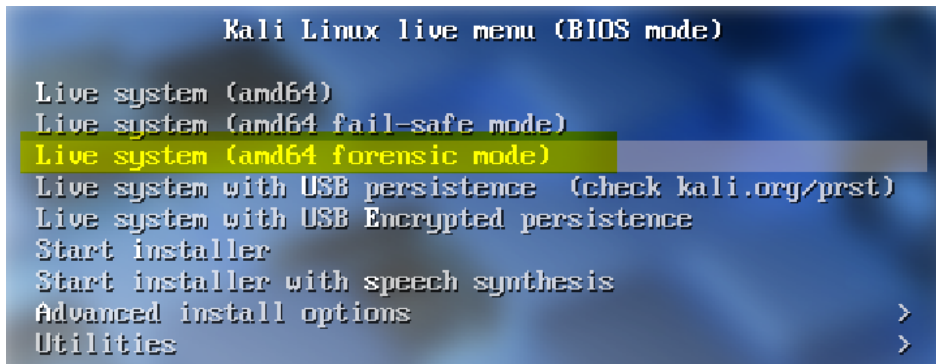


- b) Start the **Windows VM** and boot into **Kali**.

[Note: Check the boot order via VirtualBox Manager > Settings > System. All changes have to be made when the VM powered off.]



- c) In Kali boot up screen, select **Live System (amd64 forensic mode)**



- d) In order to determine the correct hard disk device to acquire, start **Terminal** console and use the command **"sudo -s; fdisk -l"**.

[Note: You should be able to determine the correct device based on the disk size.]

```
(kali@kali)-[~]
$ sudo -s
#
(root@kali)-[/home/kali]
# fdisk -l

Disk /dev/sda: 100 GiB, 107374182400 bytes, 209715200 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xc1043e91

Device      Boot  Start      End  Sectors  Size Id Type
/dev/sda1   *        2048    1187839   1185792   579M  7 HPFS/NTFS/exFAT
/dev/sda2                1187840 209711103 208523264   99.4G  7 HPFS/NTFS/exFAT

Disk /dev/sdb: 200 MiB, 209715200 bytes, 409600 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x0f533a5e

Device      Boot  Start      End  Sectors  Size Id Type
/dev/sdb1                128  403583    403456   197M  7 HPFS/NTFS/exFAT
```

- e) Do a disk-to-disk copy to acquire the disk partition **/dev/sdb** to a disk image file

```
# dd if=/dev/sdb of=/home/kali/200MB.dd status=progress bs=1M
200+0 records in
200+0 records out
209715200 bytes (210 MB, 200 MiB) copied, 0.388236 s, 540 MB/s

(root@kali)-[/home/kali]
# ls -lh /home/kali/200MB.dd
-rw-r--r-- 1 root root 200M May  5 02:42 /home/kali/200MB.dd

(root@kali)-[/home/kali]
# md5sum /home/kali/200MB.dd
867856bda5c2c05aa565e9b9e2c76c5b /home/kali/200MB.dd
```

- f) What is the MD5 hash value of the local drive **/dev/sdb**? Does its hash value tally with your acquired disk image hash value?

3. Part 3. Reconstruction from disk image to physical drive

Now, you would restore the acquired 200MB disk image file into the 300MB hard disk using Linux command line tool.

- a) Ascertain the physical disk to reconstruct to; /dev/sdc (300MB)

```
(root@kali)~# fdisk -l /dev/sdc
Disk /dev/sdc: 300 MiB, 314572800 bytes, 614400 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

(root@kali)~#
```

- b) Shred /dev/sdc to sanitize the disk

```
(root@kali)~# shred -v /dev/sdc
shred: /dev/sdc: pass 1/3 (random) ...
shred: /dev/sdc: pass 2/3 (random) ...
shred: /dev/sdc: pass 3/3 (random) ...

(root@kali)~#
```

- c) Reconstruct from the previously acquired disk image file to the physical disk

```
(root@kali)~# dd if=/home/kali/200MB.dd of=/dev/sdc status=progress bs=1M
200+0 records in
200+0 records out
209715200 bytes (210 MB, 200 MiB) copied, 0.44425 s, 472 MB/s

(root@kali)~# fdisk -l /dev/sdc
Disk /dev/sdc: 300 MiB, 314572800 bytes, 614400 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x0f533a5e

Device      Boot Start    End Sectors  Size Id Type
/dev/sdc1   128 403583 403456 197M  7 HPFS/NTFS/exFAT
```

- d) Verify the reconstructed /dev/sdc

```
(root@kali)~# mount /dev/sdc1 /mnt

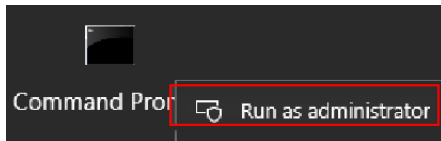
(root@kali)~# ls -l /mnt
total 1
drwxrwxrwx 1 root root  0 May  4 12:41 '$RECYCLE.BIN'
-rwxrwxrwx 1 root root 24 May  4 12:50 'MyFile.txt'
drwxrwxrwx 1 root root  0 May  4 12:39 'System Volume Information'

(root@kali)~# md5sum /mnt/MyFile.txt
0049dc7a38ce8e120684df2a61522c4f  /mnt/MyFile.txt

(root@kali)~# cat /mnt/MyFile.txt
Name:
ID Number:
Date:

(root@kali)~# umount /mnt
```

- e) Shut down Kali, remove the CD. Boot the **Windows VM** as normal.
- f) Restore the clone disk attribute



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.18363.592]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Windows\system32>diskpart

Microsoft DiskPart version 10.0.18362.1

Copyright (C) Microsoft Corporation.
On computer: DESKTOP-PP0L10Q

DISKPART> list disk

   Disk ###  Status              Size       Free       Dyn  Gpt
   -----  -
   Disk 0      Online               100 GB     1024 KB
   Disk 1      Online               200 MB     1984 KB
   Disk 2      Online               300 MB      300 MB

DISKPART> select disk 2

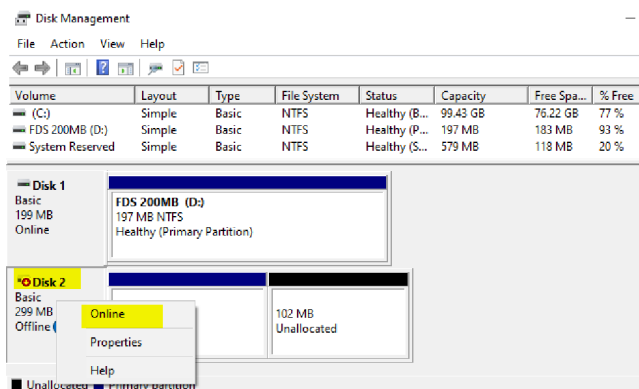
Disk 2 is now the selected disk.

DISKPART> attributes disk clear readonly

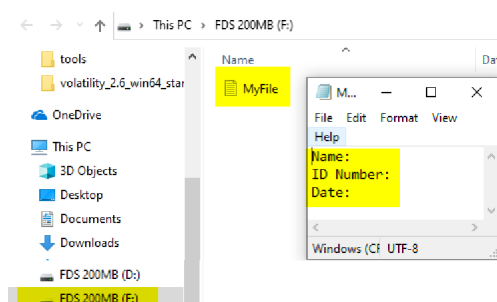
Disk attributes cleared successfully.

DISKPART> exit

Leaving DiskPart...
```



- g) Open the text file inside the reconstructed disk and show your lecturer the MD5 hash of the text file is the same for both the original (D drive) and clone disk (F drive).



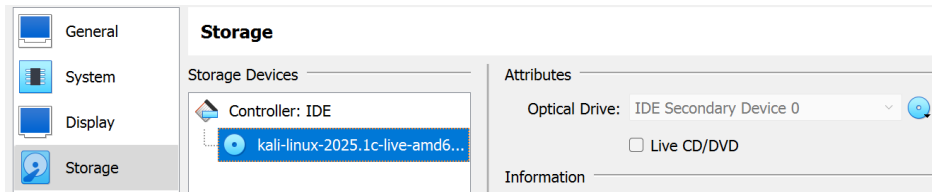
```
C:\Windows\system32>certutil -hashfile D:\MyFile.txt MD5
MD5 hash of D:\MyFile.txt:
0049dc7a38ce8e120684df2a61522c4f
CertUtil: -hashfile command completed successfully.

C:\Windows\system32>certutil -hashfile F:\MyFile.txt MD5
MD5 hash of F:\MyFile.txt:
0049dc7a38ce8e120684df2a61522c4f
CertUtil: -hashfile command completed successfully.
```

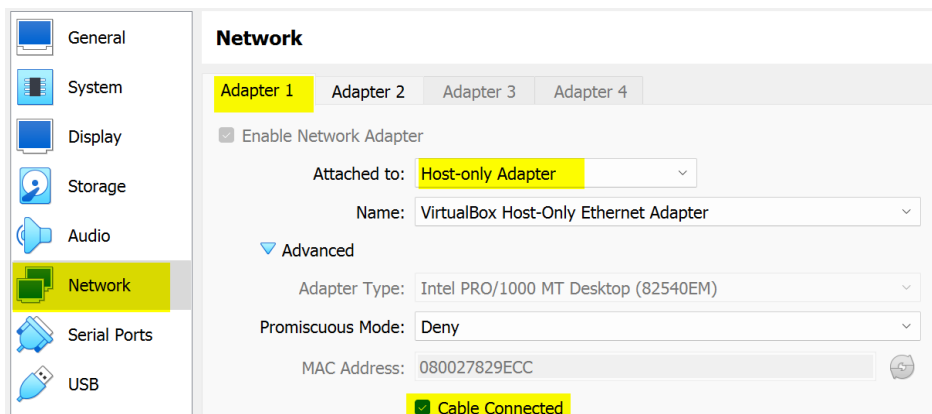
4. Part 4. Remote Acquisition

Instead of acquiring image locally on the machine, acquisition can be done remotely over the network.

- a. Create a new diskless VM with only a CDROM to run **Kali** Live CD as the remote machine.



- b. On both VMs, set the Network Adapter 1 to be Host-only Adapter with the Cable connected.



- c. Boot both the Windows VM and the new remote VM using Kali Live CD
- d. Verify both the VMs can ping each other via the Host-only Adapter 1.

```
(kali@kali)-[~]
$ ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.104 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::4b62:34a3:1722:4d68 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:82:9e:cc txqueuelen 1000 (Ethernet)
    RX packets 42 bytes 7997 (7.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 38 bytes 4912 (4.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)-[~]
$ ping -c 3 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.634 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=2.48 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=1.50 ms

— 192.168.56.103 ping statistics —
3 packets transmitted, 3 received, 0% packet loss, time 2017ms
rtt min/avg/max/mdev = 0.634/1.539/2.482/0.754 ms
```

- e. Start ssh on the remote VM

```
└─$ ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.104 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::121:3ec:d962:6c92 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:82:9e:cc txqueuelen 1000 (Ethernet)
    RX packets 107 bytes 10140 (9.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 100 bytes 10644 (10.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali㉿kali)-[~]
└─$ sudo -s
(kali㉿kali)-[/home/kali]
└─# systemctl start ssh

(kali㉿kali)-[/home/kali]
└─# systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: active (running) since Mon 2025-05-05 04:40:29 UTC; 9s ago
```

- f. Do a disk-to-disk copy from Windows VM booted in Kali to the remote Kali VM via ssh. The ssh password is "kali"

```
└─# dd if=/dev/sdb status=progress bs=1M | ssh kali@192.168.56.104 dd of=/home/kali/200MB_remote.dd
kali@192.168.56.104's password: kali
179306496 bytes (179 MB, 171 MiB) copied, 7 s, 25.6 MB/s
200+0 records in
200+0 records out
209715200 bytes (210 MB, 200 MiB) copied, 7.79615 s, 26.9 MB/s
409600+0 records in
409600+0 records out
209715200 bytes (210 MB, 200 MiB) copied, 4.93126 s, 42.5 MB/s
```

- g. Verify the hash value of the local drive (/dev/sdb) should be the same as the remotely acquired disk image file.

```
└─# ssh kali@192.168.56.104 "md5sum /home/kali/200MB_remote.dd"
kali@192.168.56.104's password:
867856bda5c2c05aa565e9b9e2c76c5b /home/kali/200MB_remote.dd

(kali㉿kali)-[/home/kali]
└─# md5sum /dev/sdb
867856bda5c2c05aa565e9b9e2c76c5b /dev/sdb
```

What is the advantage(s) of using this method of imaging from the victim's machine to a remote machine? What is the disadvantage of this method of image transfer?

END